

AI EPC Platform - Engineering Specification

Engineering Specification

Project : Beta Solar Plant

Vendor : Siemens

Equipment : Pump

Issue Date : 2025-01-12

Project Overview

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The project consists of the construction of a Tier III data center supporting mission-critical operations.

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Cooling System

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CRAH units shall support automatic failover.

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Chilled water pumps shall operate with redundant motors.

Cooling equipment shall support SNMP monitoring.

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Cooling redundancy shall be verified during commissioning.

Each cooling unit shall include vibration monitoring sensors.

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The cooling system shall provide N+1 redundancy.

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Electrical System

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Generator backup shall support 72 hours of continuous operation.

Dual power feeds shall be provided for critical equipment.

Power Distribution Units shall provide branch circuit monitoring.

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Emergency shutdown shall disconnect non-critical loads.

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UPS capacity shall be 500 KVA.

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Fire Protection

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FM200 suppression shall protect all server rooms.

Emergency evacuation procedures shall be displayed.

Smoke detectors shall be installed in all electrical rooms.

Fire alarm panels shall integrate with the BMS.

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Network Infrastructure

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Fiber optic backbone shall use single-mode fiber.

Network monitoring shall integrate with the NOC.

Dual fiber entry points shall be provided.

All network racks shall use Cat6A cabling.

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Generator synchronization tests shall be completed.

All UPS systems shall pass load testing.

Commissioning reports shall include all test results.

Cooling performance shall be verified under full load.

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Quality Assurance

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